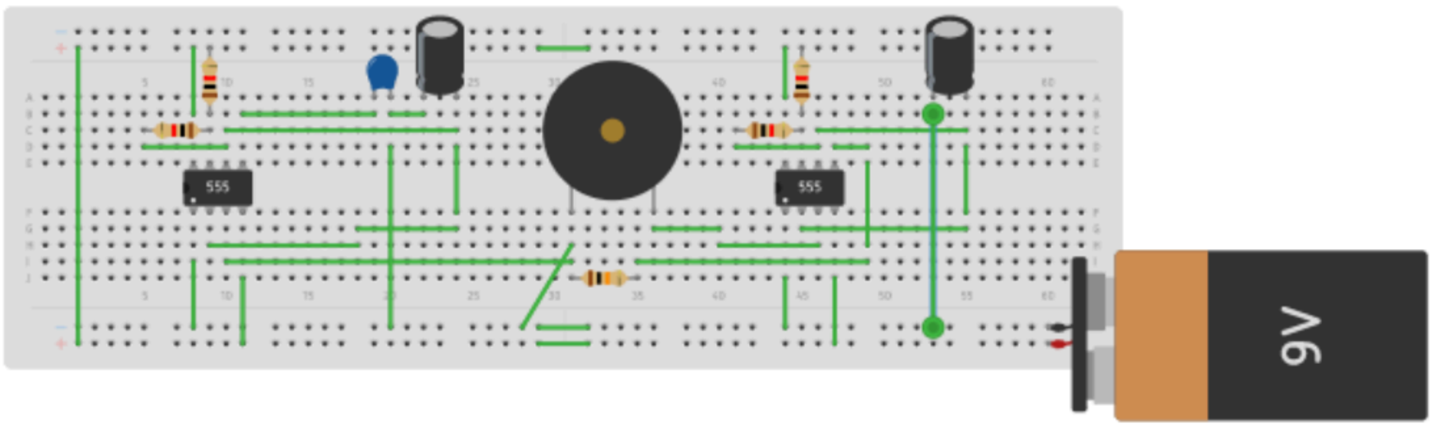
MINI ELECTRONICS



TEACHER: CERTIFIED TRAINER	DURATION: 16 CLASSES (1 HOUR PER CLASS)	MODE: ONLINE

This is a virtual, hands-on mini electronics curriculum designed for early learners using Tinkercad as the learning platform. Students build and simulate electronic circuits using drag-and-drop components such as LEDs, buzzers, switches, sensors, motors and ICs in a safe digital environment. Each session involves creating functional circuits to understand real-world applications while learning core concepts like current, voltage, resistance, switches, sensors and basic logic. Tinkercad enables easy experimentation, instant results, and error-free learning, encouraging curiosity, creativity and problem-solving skills.

COURSE CURRICULUM

1. Glow Control Basics

- Introduction to electronics and its importance in daily life.
- Getting familiar with Tinkercad interface and basic electronic components such as LED, battery and resistor.

2. Sound Alert System

- Understanding how a buzzer works.
- Creating a simple sound alert circuit using a buzzer in Tinkercad.

3. Press-to-Glow Circuit

- Learning the function of a push button.
- Building a circuit where an LED turns ON when the button is pressed.

4. Slide Switch Operations

- Understanding ON/OFF control using a slide switch.
- Creating circuits to control power flow with a slide switch.

5. Light and Sound Effects

- Combining LED and buzzer in a single circuit.
- Using a slide switch to control light and sound together.

6. Traffic Light System

- Understanding traffic signal logic.
- Building a basic traffic light circuit using LEDs and push buttons.

7. Level Control Circuit

- Introduction to potentiometer as a variable resistor.
- Controlling LED brightness and buzzer volume using a potentiometer.

8. Multi-Color Light System

- Understanding RGB LED and color mixing.
- Creating colorful light effects using RGB LEDs.

9. Water Level Alert Circuit

- Introduction to the transistor as a switch.
- Building a water level indicator circuit using a transistor.

10. Automatic Light Sensor

- Understanding Light Dependent Resistors (LDR).
- Building a circuit that responds to light and darkness.

11. Motor Speed Regulator

- Introduction to DC motor operation.
- Controlling motor speed using electronic components.

12. Energy Storage Behavior

- Understanding capacitors and energy storage.
- Learning charging and discharging behavior through simulation.

13. Timing and Pulse Generator

- Introduction to the 555 Timer IC.
- Creating simple timing and blinking circuits.

14. Temperature Sensing Device

- Understanding thermistor and temperature sensing.
- Building a temperature-responsive circuit.

15. Digital Decision Circuits

- Introduction to logic gates (AND, OR, NOT).
- Understanding digital inputs, outputs and decision-making circuits.

16. Challenge Day

- Problem-solving challenges using Tinkercad circuits.
- Testing understanding, creativity and troubleshooting skills through mini tasks and projects.

Class Structure

1) Introduction to Electronic Components

- Learn about basic electronic components and their functions using virtual components in Tinkercad.

2) Virtual Circuit Assembly (Tinkercad-Based)

- Design and assemble electronic circuits using drag-and-drop components in a safe digital environment.

3) Concept Understanding Through Simulation

- Understand electronic principles through real-time simulation and clear visual feedback linked to real-life devices.

4) Hands-on Activities and Mini Projects

- Create interactive circuits, experiments, and challenges to reinforce learning and problem-solving skills.